KT1

Ruhr-Universität Bochum
Experimental Physics I

WINTER 2026/27 COURSE 160202

Introduction to Nuclear and Particle Physics

Exercise sheet 1

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Assistant Dr. Marian Stahl
Tutors Hendric Jonas
Dhruvanshu Parmar
Due October 23, 2026

Textbook. Mark Thomson, *Modern Particle Physics*, Cambridge University Press, first edition (2013).

This sheet assigns the exercises by reference. Use the problem statements and figures in your copy of the textbook.

Exercise 1 — Thomson 1.1

Ref.: Lecture 1

Exercise 2 — Thomson 1.4

Ref.: Lecture 1 + chapter 1

Exercise 3 — Thomson 1.10

Ref.: Lecture 2

Exercise 4 — Thomson 2.16

Ref.: QM course + chapter 2

Submission and preparation for the tutorial

Write your solutions by hand and work individually. Please hand your solutions to your tutor by October 23. AI assistance is permitted for preparation; be ready to explain your reasoning independently. Bring one question or an unfinished attempt to discuss with your tutor.